Lecture 8:
Unsupervised Machine Learning and Kmeans clustering

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Agenda

- Revision of train test split and working of supervised learning
- Unsupervised Machine Learning
- Types of Unsupervised Machine Learning
- Kmeans clustering

Supervised Learning

	loan_id	income_annum	loan_amount	loan_term	cibil_score	residential_assets_value	bank_asset_value	loan_status
0	1	9600000	29900000	12	778	2400000	8000000	Approved
1	2	4100000	12200000	8	417	2700000	3300000	Rejected
2	3	9100000	29700000	20	506	7100000	12800000	Rejected
3	4	8200000	30700000	8	467	18200000	7900000	Rejected

R&D Spend	Administration	Marketing Spend	State	Profit
165349.2	136897.8	471784.1	New York	192261.83
162597.7	151377.59	443898.53	California	191792.06
153441.51	101145.55	407934.54	Florida	191050.39
144372.41	118671.85	383199.62	New York	182901.99
142107.34	91391.77	366168.42	Florida	166187.94
131876.9	99814.71	362861.36	New York	156991.12

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Train-test split and supervised learning

X features

y target

	Hours Studied	Previous Scores	Extracurricular Activities	Sleep Hours	Sample Question Papers Practiced	Performance Index
0	7	99	1	9	1	91.0
1	4	82	0	4	2	65.0
2	8	51	1	7	2	45.0
3	5	52	1	5	2	36.0
4	7	75	0	8	5	66.0
...	...	...	...	...	...	...
9995	1	49	1	4	2	23.0
9996	7	64	1	8	5	58.0
9997	6	83	1	8	5	74.0
9998	9	97	1	7	0	95.0
9999	7	74	0	8	1	64.0

X_TRAIN

y_train

X_TEST

y_test

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Classroom with No Teacher

A class where:

No teacher gives grades, no labels, no instructions.
The students simply **sit**, **behave**, **talk**, and the system has to **figure out patterns** on its own.

This is unsupervised ML

No one told me what is right or wrong.
But I will find patterns on my own.

Types of Unsupervised Learning:

- **Clustering**
- **association rule learning**
- **dimensionality reduction**

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How Netflix figure out which groups of users prefer action movies, comedi dramas?

Supermarkets: which produ bought together

group animals into categ mammals, birds, and reptil

How cities decide the best for new hospitals, fire stations

Divide people into clusters

K Means clustering comes into action

Clustering, finding natural groupings

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Similarity and clusters

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K Means Clustering

You are organizing some party

Different types of people will be there

You observe and find the interest of them

No One tells you who is of what type

Music

Studious

Freak teams

Sports interested

You draw imaginary circles around groups of similar behavior.

That's K-Means:
K = number of groups
Means = center of each group (centroid)

Algorithm speaking:
"I will assign each person to the nearest group and then adjust the group cente
then repeat until everyone is happy."

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Hierarchical Clustering – "The Family Tree Maker"

Think of family tree:
First: everyone is alone
Then: cousins join together
Then: families merge
Then: bigger families merge
Finally you get a full tree.

Hierarchical clustering works the same:
Start with every data point as a separate cluster
Merge the two closest groups
Keep merging until everything becomes one cluster

PCA – "The Magician Who Shrinks Without Losing Meaning"

Imagine your room is messy with objects all over the floor.
Your mom says:
"Clean it but don't throw anything important!"

You:
Compress scattered items
Organize them in the corner
Keep only meaningful directions

This is **PCA (Principal Component Analysis)**:
It reduces **dimensions**
But keeps **maximum variance**
Like turning a 100-column dataset into 2 or 3 columns while keeping meaning

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DBSCAN – “The Food Court Grouping Logic”

Go to a food court.
People naturally form groups:
Pizza corner crowd
Momo lovers
Coffee drinkers
DBSCAN says:
Wherever people crowd together = a cluster
A person sitting alone = noise

Autoencoders – “The Data Copy Machine That Learns by Compressing”

Imagine a magical photocopy machine:
It sees a file
Squashes it into a tiny compressed version
Then reconstructs the full file
If it learns to compress well → it understands patterns
If reconstruction is bad → the file is an anomaly

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Hierarchical Clustering	PCA	DBSCAN	Autoencoders
Produces dendrogram No need to choose K initially	PCA Component = Eigenvector Component Importance = Eigenvalue Useful for: Compression Noise reduction Visualization	Pros: Finds arbitrary-shape clusters Handles noise Cons: Sensitive to ε(radius) selection	Encoder → compress Decoder → reconstruct Loss → reconstruction error Variants: Denoising Variational (VAE) Sparse

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K-Means Clustering

- One of the simplest algorithms for clustering
- Based on **Euclidean Distance**
- Divides data in into K groups (clusters)
- **K-means clustering**
- unsupervised machine learning algorithm used to partition a dataset into distinct groups (or "clusters") based on the similarities of data points.
- to discover patterns, group similar items, or simplify data into representative clusters without any predefined labels.

Usage

- **Grouping Similar Data:**
  - K-means groups data points that are close to each other into clusters, so each cluster contains data points with similar characteristics.
- **Pattern Discovery:**
  - In the absence of labeled data, K-means helps identify underlying structures or patterns within the dataset.

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Working

- **Step 1**: ***Choose the Number of Clusters K***
  - Before starting, decide how many clusters K you want the algorithm to create. This number is often based on domain knowledge or by experimenting.
- **Step 2**: ***Initialize Cluster Centroids***
  - The algorithm selects K random points in the data as initial "centroids" (centers of clusters).
- **Step 3**: ***Assign Data Points to the Nearest Centroid***
  - Each data point is assigned to the cluster with the nearest centroid based on distance (usually Euclidean distance). This forms initial clusters.

Step 4: ***Update Centroids***
For each cluster, the algorithm recalculates the centroid by taking the mean of all data points assigned to that cluster.

Step 5: *Repeat Steps 3 and 4*
The assignment and updating steps are repeated until the centroids no longer move significantly, or a set number of iterations is reached.

Final Output: The algorithm returns the clusters, with each data point assigned to a particular cluster and each cluster defined by its final centroid.

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Core Concepts in K-Means Clustering

- **Centroids**:
- Each cluster has a centroid, representing the "center" of that cluster. *A centroid is the average position of all data points within a cluster.*
- **Inertia** (**Within-Cluster Sum of Squares**):
- Inertia measures how close data points are to their respective centroids. The goal is to minimize inertia, so the data points in each cluster are as close to their centroid as possible.
- **Convergence**:
- The algorithm converges (stops) when the centroids no longer change significantly, meaning the clusters are stable.

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Unsupervised Learning

- Training data: “examples” $\boldsymbol{X}$

$$\boldsymbol{x_1, \dots, x_n}, \quad \boldsymbol{x_i} \in \boldsymbol{R^d}$$

- Clustering

$$\boldsymbol{X} \rightarrow \{\boldsymbol{C_1}, \dots, \boldsymbol{C_k}\}\ (set\ of\ clusters)\ ,$$

where $\boldsymbol{k} = number\ of\ clusters$

example: Find clusters in the population, fruits, text

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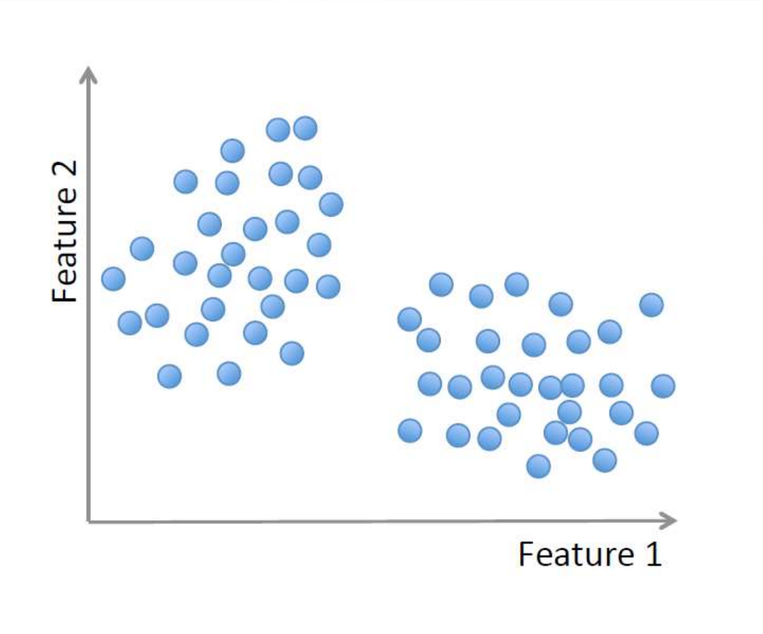
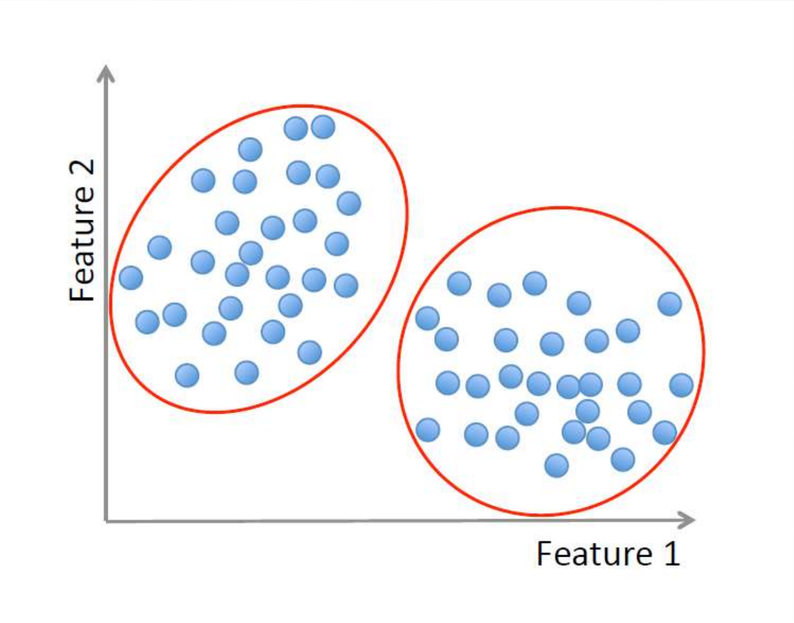
Unsupervised Learning





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Given datapoints

Given Data Points: **(sleep hr and study hr)**

A = (5, 2)

B = (6, 1)

C = (6, 2)

D = (6, 3)

E = (10, 0)

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Given datapoints

Given Data Points: **(sleep hr and study hr)**

(5, 2)
(6, 1)
(6, 2)
(6, 3)
(10, 0)

Centroid 1 (C1): (5, 2)
Centroid 2 (C2): (6, 1)

Compute Euclidean Distance of Each Point to Each Centroid

Point A = (5,2)

*Let's choose **K = 2 clusters***

*We will start with **initial centroids** using the first two points:*

To C1 (5,2): $d = \sqrt{(5-5)^2 + (2-2)^2} = 0$

To C2 (6,1): $d = \sqrt{(6-5)^2 + (1-2)^2} = \sqrt{1+1} = \sqrt{2} = 1.41$

Assigned to Cluster 1

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Given datapoints

Given Data Points: **(sleep hr and study hr)**

(5, 2)
(6, 1)
(6, 2)
(6, 3)
(10, 0)

Centroid 1 (C1): (5, 2)
Centroid 2 (C2): (6, 1)

Compute Euclidean Distance of Each Point to Each Centroid

Point B = (6,1)

To C1 (5,2): $d = \sqrt{(6-5)^2 + (1-2)^2} = 1.41$

To C2 (6,1): $d = 0$

Assigned to Cluster2

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Given datapoints

Given Data Points: **(sleep hr and study hr)**

(5, 2)
(6, 1)
(6, 2)
(6, 3)
(10, 0)

Centroid 1 (C1): (5, 2)
Centroid 2 (C2): (6, 1)

Compute Euclidean Distance of Each Point to Each Centroid

Point C = (6,2)

To C1 (5,2):

$$d = \sqrt{(6-5)^2 + (2-2)^2} = 1$$

To C2 (6,1):

$$d = \sqrt{(6-6)^2 + (2-1)^2} = 1$$

Tie: Can choose either.
We typically assign to the **nearest with lowest index**, so **Cluster 1**.

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Given datapoints

Given Data Points: **(sleep hr and study hr)**

(5, 2)
(6, 1)
(6, 2)
(6, 3)
(10, 0)

Centroid 1 (C1): (5, 2)
Centroid 2 (C2): (6, 1)

Compute Euclidean Distance of Each Point to Each Centroid

Point D = (6,3)

To C1 (5,2): $d = \sqrt{(6-5)^2 + (3-2)^2} = \sqrt{1+1} = 1.41$

To C2 (6,1): $d = \sqrt{0^2 + 2^2} = 2$

Assigned to **Cluster 1**

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Given datapoints

Given Data Points: **(sleep hr and study hr)**

(5, 2)
(6, 1)
(6, 2)
(6, 3)
(10, 0)

Centroid 1 (C1): (5, 2)
Centroid 2 (C2): (6, 1)

Compute Euclidean Distance of Each Point to Each Centroid

Point E = (10,0)

To C1 (5,2): $d = \sqrt{(10-5)^2 + (0-2)^2} = \sqrt{25+4} = \sqrt{2}$

To C2 (6,1): $d = \sqrt{(10-6)^2 + (0-1)^2} = \sqrt{16+1} = \sqrt{}$

Assigned to **Cluster 2**

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Clusters after Iteration 1

Cluster 1:
A = (5,2)
C = (6,2)
D = (6,3)

Cluster 2:
B = (6,1)
E = (10,0)

Step 2: Update the Centroids
New Centroid 1 (mean of C1 points)
Points: (5,2), (6,2), (6,3)
Compute average:
x=(5+6+6)/3=17/3=5.67
y=(2+2+3)/3=7/3=2.33

C1' = (5.67, 2.33)

New Centroid 2 (mean of C2 points)
Points: (6,1), (10,0)
x=(6+10)/2=8
y=(1+0)/2=0.5

C2' = (8, 0.5)

Iteration 2 (Optional)
If we run distances again, you'll find:
Cluster assignments **do not change**
So algorithm **converged** after Iteration 1.

Cluster 1 (Sleep-Study Balanced Group):
(5,2)
(6,2)
(6,3)

Cluster 2 (Long Sleep–Low Study Group):
(6,1)
(10,0)

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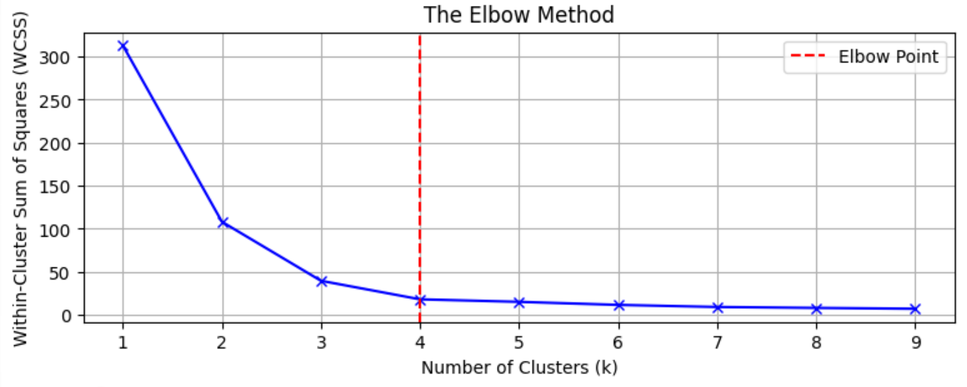
Determining Optimal Number of Clusters (K)

- **Elbow Method:**
- This method involves running K-means clustering with different values of K and plotting the inertia. *The "elbow point" in the plot (where the inertia decreases sharply and then levels off) suggests an optimal K.*
- **Silhouette Score**:
- The silhouette score measures how similar each data point is to its own cluster compared to other clusters. *A higher score indicates a better-defined clustering structure.*



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Applications of K-Means Clustering

Customer Segmentation:

- segment customers based on purchasing behavior, demographics, or other attributes to tailor their strategies.

Image Compression:

- By clustering pixels in an image, K-means can reduce the number of colors used, thereby compressing the image size.

Anomaly Detection:

- In fraud detection, normal and abnormal behaviors can be clustered. Points that don't belong to any cluster are flagged as anomalies.

Document Classification:

- K-means can be used to cluster similar documents or articles based on word frequency or content, even without predefined categories.

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Advantages and Disadvantages

Advantages:

- **Simple and Easy to Implement:**
  - K-means is straightforward and computationally efficient, especially with large datasets.
- **Fast and Scalable:**
  - It can handle large amounts of data by partitioning it into manageable clusters.

Disadvantages:

-Choosing K Can Be Challenging:
It requires setting the number of clusters in advance, which might not be obvious without domain knowledge.

Sensitive to Initialization:
If initial centroids are poorly chosen, K-means may converge to a suboptimal solution. Using techniques like K-means++ can help select better initial centroids.

Assumes Spherical Clusters:
K-means works best with clusters that are circular or spherical. It may not perform well with irregular or overlapping clusters.

Sensitive to Outliers:
Outliers can distort centroids, leading to inaccurate clustering.

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Example Scenario

- A retail company wants to segment its customers to better target marketing campaigns. They gather data on each customer's purchase history, like frequency of purchase, total spending, and types of products bought.

Step 1:

- **choose k=3 clusters, assuming they want three customer segments** (e.g., "High-Spenders," "Moderate-Spenders," and "Low-Spenders").

Step 2:

- The algorithm initializes three centroids randomly in the dataset.

Step 3:

- Each customer is assigned to the nearest centroid based on their purchase characteristics.

Step 4:

- The centroids are recalculated based on the assigned customers, and the assignments are updated.

Step 5:

- Steps 3 and 4 repeat until the clusters stabilize, resulting in three distinct customer groups.

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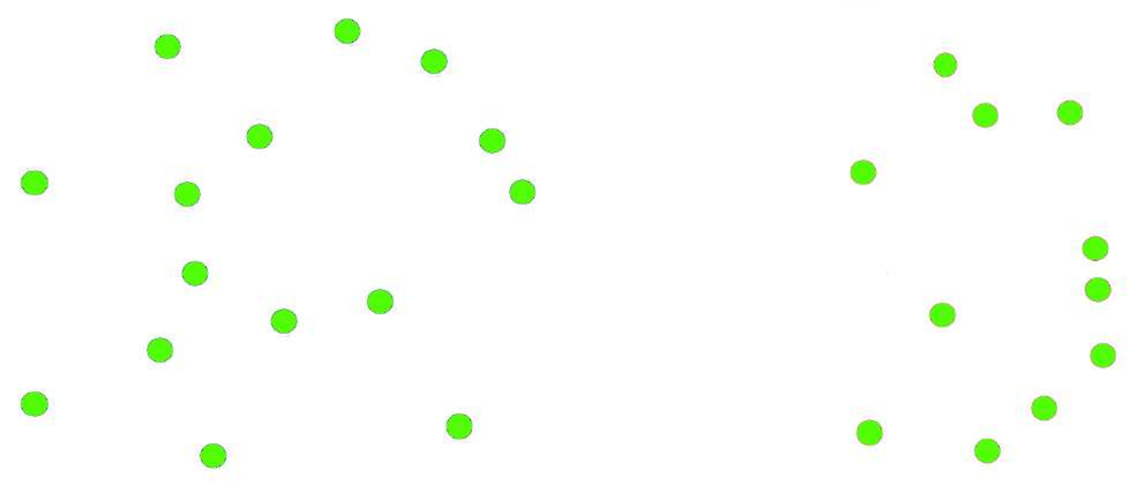
K-Means Clustering

given:
data points
with no labels

goal:
divide data
points into 2
groups



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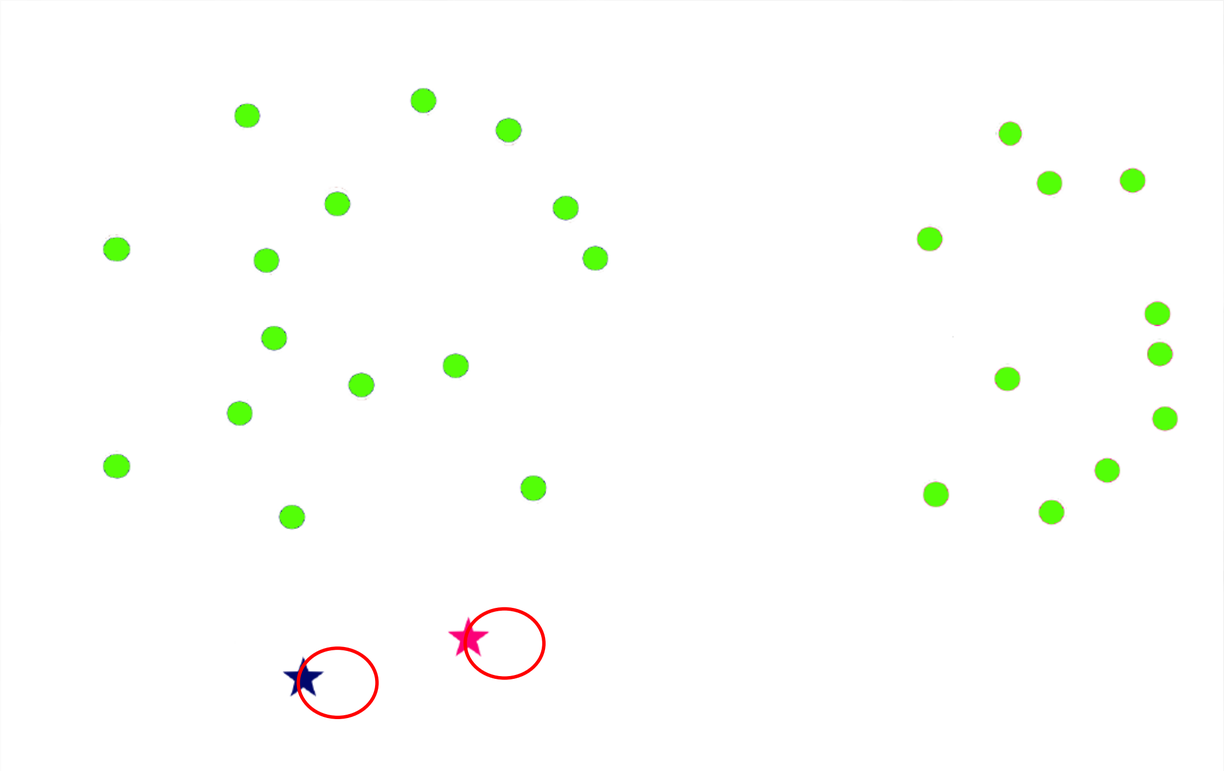
K-Means Clustering

- *randomly drop 2 points in the feature space*
- *we will call them the centroids of the clusters*



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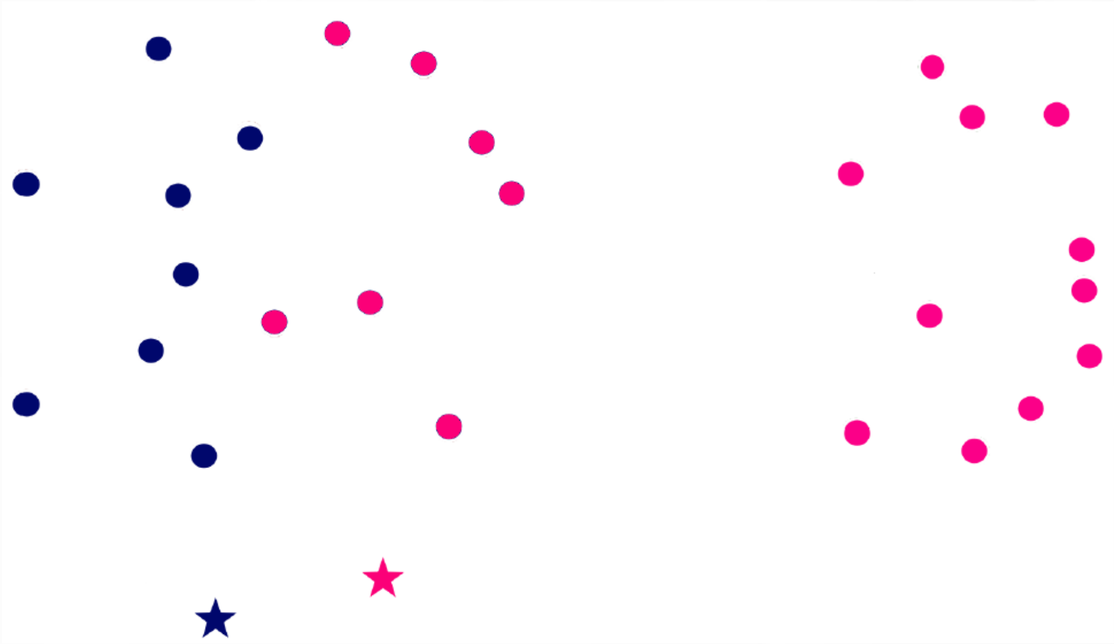
K-Means Clustering



- *calculate the distance between the 2 centroids and every other point*
- *assign each point to the cluster whose centroid is closest to the point*

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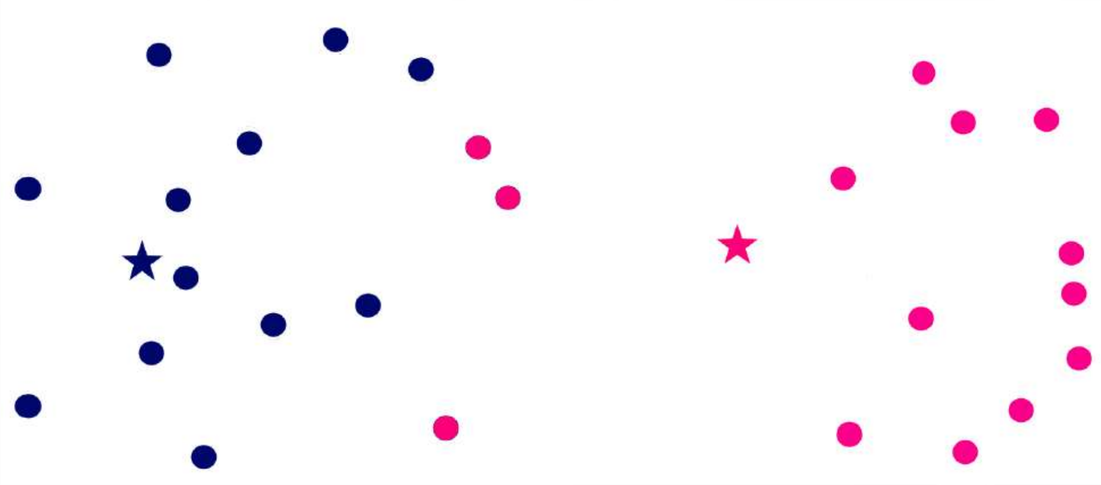

- *calculate the new centroids for the 2 clusters by taking the average/mean of all the data points in respective clusters*
- *make new cluster assignments using the updated centroids*

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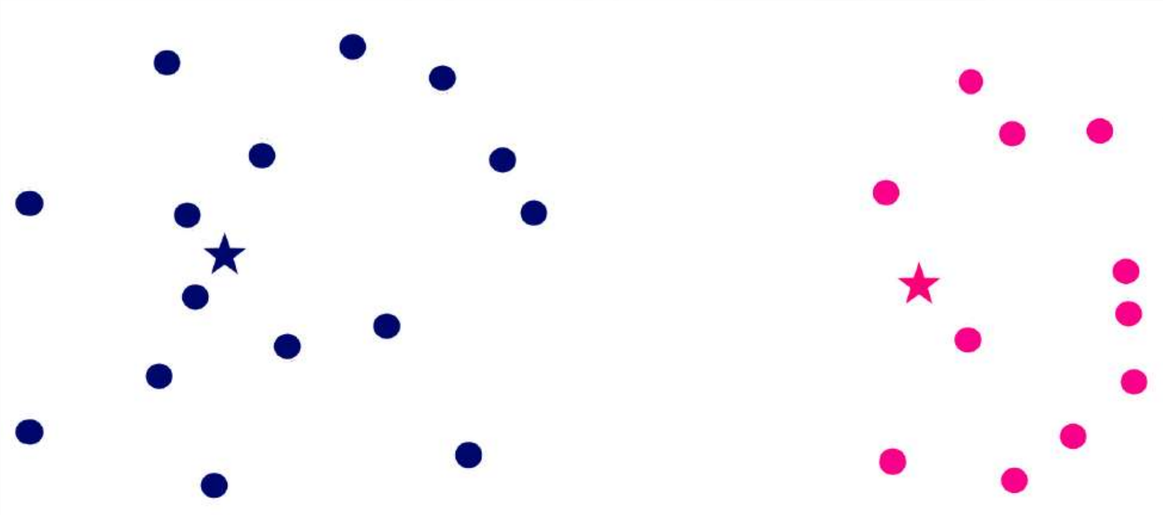

- *repeat the same process until the values for the centroids stop changing/no change in the clusters*

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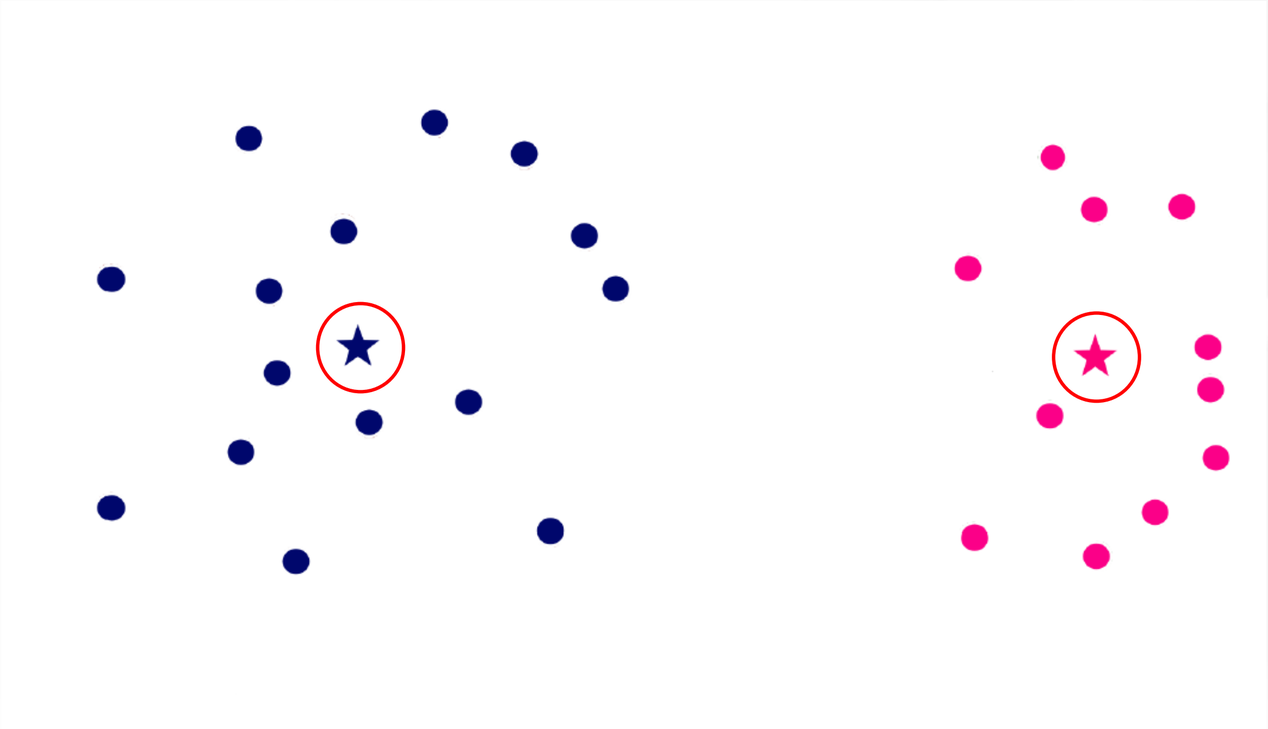

- *the data points have been divided into 2 groups/clusters with their respective centroids*

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Thank you !

End of Lecture 8